

Well-Written Proofs

There are three main things that I look for in any well-written proof. Your proof should be

1. Clear
2. Concise
3. Convincing

Note that “Correct” is not one of the criteria—because it should follow from the previous three items! Here are a few tips and miscellaneous advice which you can use while writing your proofs. This list is by no means exhaustive!

- Start every sentence with a capital letter and a punctuation mark. In particular, do not begin a new sentence with an equation.

Not C/C/C: Suppose a and b are integers with $a \mid b$. $b = a \cdot k$ for some integer k .

C/C/C: Suppose a and b are integers with $a \mid b$. Then $b = a \cdot k$ for some integer k .

- Do not say the word “it” when referencing something previously introduced. No one knows what “it” refers to unless you take extra time to clarify what “it” means, and at that point “it” is more of a hindrance than a help.

Not C/C/C: Suppose a and b are integers with $a \mid b$. It is $a \cdot k$ for some integer k .

C/C/C: Suppose a and b are integers with $a \mid b$. Then $b = a \cdot k$ for some integer k .

- Every variable must be appropriately introduced, either as something which comes from other symbols already in the picture, or as a consequence of definitions/theorems. The words “let” or “define” can be used to signal to the reader that you are introducing a new variable.

Not C/C/C: Suppose $a \mid b$. Then $b = a \cdot k$.

C/C/C: Suppose a and b are integers with $a \mid b$. Then $b = a \cdot k$ for some integer k .

- Do not commit the fallacy of “equivocation”. Once you have given a symbol a certain meaning, do not change the meaning of that symbol partway through your argument. For example consider the following **Not C/C/C** argument:

Suppose that a , b , and c are integers, and that $a \mid b$. Then $b = a \cdot k$ where k is some integer. Therefore,

$$b \cdot c = (a \cdot k) \cdot c = a \cdot k$$

so that $a \mid b \cdot c$.

In the final equality, the symbol k has been re-purposed to mean the product $k \cdot c$. Implicitly hidden in this is the idea we could have written $k = k \cdot c$ which then tells us that $c = 1$, which isn’t necessarily true. Obviously that isn’t what is intended, but it opens the door for that misinterpretation, and is **Not C/C/C** and thus is misleading.

Instead, look at the following **C/C/C** argument:

Suppose that a , b , and c are integers, and that $a \mid b$. Then $b = a \cdot k$ where k is some integer. Let $k' = k \cdot c$, which is an integer as a product of integers. Then

$$b \cdot c = (a \cdot k) \cdot c = a \cdot k'.$$

Therefore, $a \mid b \cdot c$.

- Logical connectives can only connect *statements*. In particular, they cannot connect numbers (though they can connect equations). However, you can use strings of equality signs to show that two things are equal. In

the following assume we already know $n = 2k$.

Not C/C/C: $n^2 \implies (2k)^2 \implies 4k^2 \implies 2(2k^2)$.

C/C/C: $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$.

- Use filler words such as “therefore”, “thus”, “so then”, “however”, “hence”, “which implies”, etc., to connect your sentences. Even if your math is being aligned in the center of the page, there shouldn’t be any “floating math”.

Not C/C/C: Suppose $n = 2k$ for some integer k .

$$n^2 = (2k)^2 = 2(2k^2).$$

C/C/C: Suppose $n = 2k$ for some integer k . Then

$$n^2 = (2k)^2 = 2(2k^2).$$

- If you want partial credit on any proof but can’t figure out how the proof goes...
 - ... start your proof with “Suppose...” and state all the initial assumption(s).
 - ... end your proof with “Therefore...” and state what has been concluded.